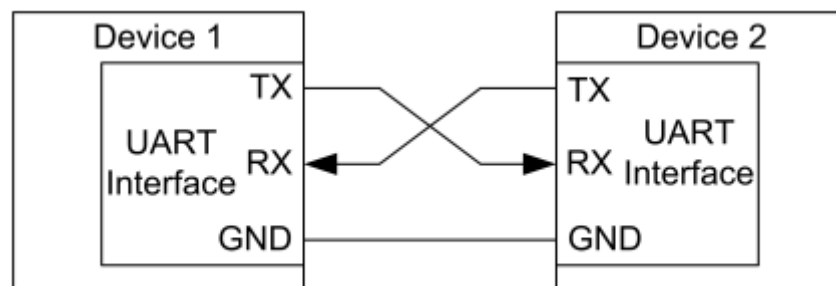


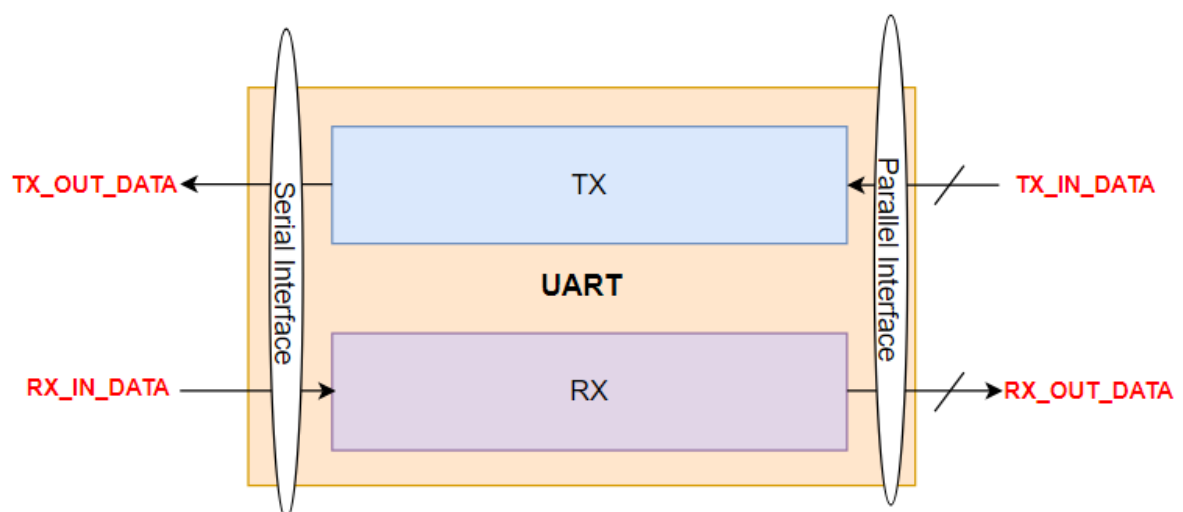
UART Transmitter

Introduction: -

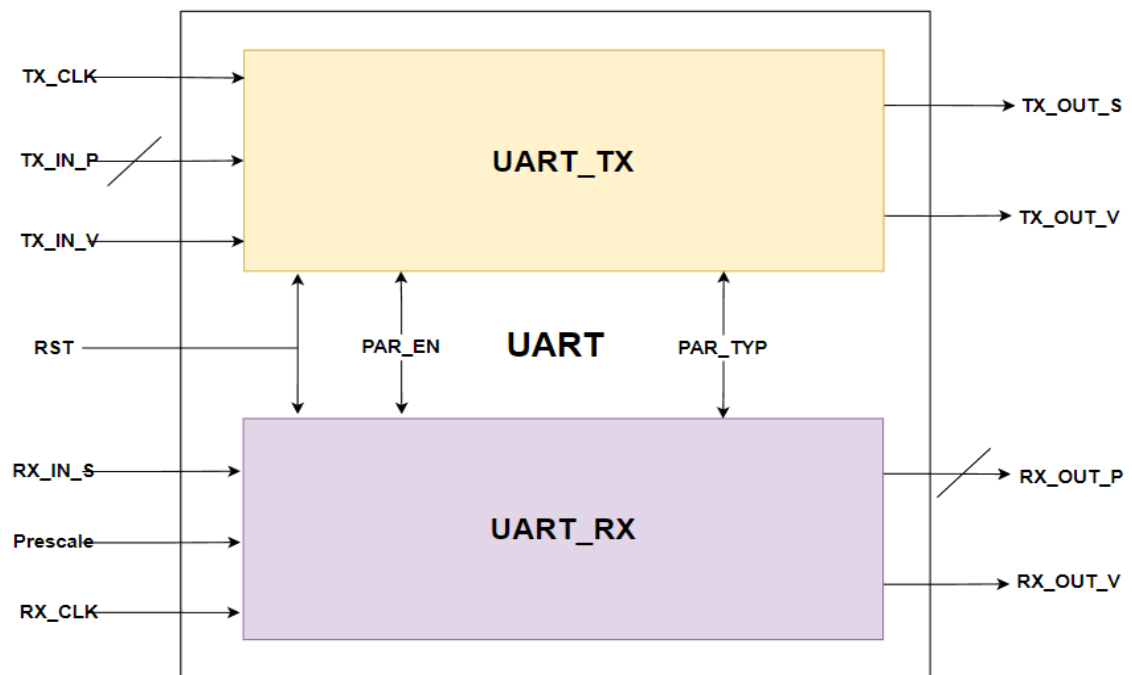
- There are many serial communication protocols as I2C, UART and SPI.
- A **U**niversal **A**synchronous **R**eceiver/**T**ransmitter (UART) is a block of circuitry responsible for implementing serial communication.
- UART is Full Duplex protocol (data transmission in both directions simultaneously)



- **Transmitting UART** converts parallel data from the master device (eg. CPU) into serial form and transmits in serial to the receiving UART.
- **Receiving UART** will then convert the serial data back into parallel data for the receiving device.



Block Interface: -



Port	Width	Description
TX_CLK	1	UART TX Clock Signal
RX_CLK	1	UART RX Clock Signal
RST	1	Synchronized reset signal
PAR_TYP	1	Parity Type
PAR_EN	1	Parity_Enable
Prescale	6	Oversampling Prescale
TX_IN_P	8	Input TX data byte
TX_IN_V	1	Input TX data valid signal
TX_OUT_S	1	TX Frame Serial Out
TX_OUT_V	1	TX Out Valid signal
RX_IN_S	1	Input RX UART frame
RX_OUT_P	8	RX Out Data
RX_OUT_V	1	RX Out Data Valid signal

Requirements: -

- 1- Integrate UART_TX & UART_RX under a top module “UART”
- 2- Write a testbench to validate that UART_TX and UART_RX are working simultaneously (Full Duplex Protocol)